

CMU 21-228

Discrete Mathematics — Problem Set

Dirichlet's Approximation Theorem · Pigeonhole Principle · Monotone
Subsequences

Based on the lecture by Po-Shen Loh

Designed for internalization — not passive reading

Study method: Slate first. Book after. Always.

How to Use This Problem Set

Rule 1: Never read the problem and immediately write an answer. Sit with it for 2 minutes first.

Rule 2: All computation on your slate. Not paper. Not phone. Slate only.

Rule 3: When stuck — do not look at a hint for at least 10 minutes. The discomfort is the rep.

Rule 4: After each block — erase everything. Reconstruct from memory before moving on.

Schedule

When	Problems
Tonight	Block 1 (all) + Block 3 — Problems 3.1, 3.2
Tomorrow morning (after gym)	Block 2 (all) + Block 3 — Problem 3.3
Tomorrow evening	Block 4 + Block 5 (Connection Journal)
Every night this week	End-of-Day Ritual

BLOCK 1 — Feel the Number Circle First

Before any theorem, just play. Your goal here is geometric feeling, not algebraic correctness.

Problem 1.1 — Place Points on the Circle

Take $\alpha = 0.37$. Draw a circle on your slate. Place these points by hand:

$1\alpha, 2\alpha, 3\alpha, 4\alpha, 5\alpha, 6\alpha, 7\alpha, 8\alpha$

Then answer on the slate:

- Which point is closest to an integer?
- What is the distance from that point to the nearest integer?
- Did you need a formula to find it — or did drawing reveal it?

Hint: multiply each by 0.37, take only the decimal part, plot it on the circle.

Problem 1.2 — Turn Back the Clock

Take $\alpha = \sqrt{2} \approx 1.4142$. Compute and place on your circle:

$1\alpha, 2\alpha, 3\alpha, 4\alpha, 5\alpha$

Now 'turn back the clock' — subtract the smaller multiple from the larger one whenever two points are close together.

Write down: which difference gives you a multiple of α closest to an integer?

This is the core move of Dirichlet's proof. Feel it geometrically before you see it algebraically.

Problem 1.3 — The Ramanujan Problem

Before any calculation — just stare at $\alpha = 1/7$.

Predict: what will happen when you place 1α through 7α on the circle?

Write your prediction on the slate. Then compute and check.

Why does this case behave completely differently from $\sqrt{2}$?

Key question: what is special about rational α vs irrational α on the number circle?

BLOCK 2 — The Green Theorem in Your Hands

Make the theorem compute, not just exist. Run the full proof on specific numbers.

Problem 2.1 — Find the Approximation for π

$$\alpha = \pi \approx 3.14159..., N = 5$$

The Green Theorem guarantees there exists p/q with $1 \leq q \leq 5$ such that:

$$|\pi - p/q| \leq 1 / (q \times 5)$$

Your job: compute $|q \cdot \pi - \text{nearest integer}|$ for $q = 1, 2, 3, 4, 5$ by hand. Which q works?

After finding it — write in one sentence why the Pigeonhole Principle guaranteed this before you computed.

Expected answer: $q = 7$ gives $|7\pi - 22| \approx 0.0088$ which is less than $1/35 \approx 0.0286$. ✓

Problem 2.2 — Run the Full Proof for $\sqrt{3}$

$$\alpha = \sqrt{3} \approx 1.7320..., N = 7$$

Step 1: Divide the unit circle into 7 equal intervals of width $1/7$.

Step 2: Place $0\alpha, 1\alpha, 2\alpha, 3\alpha, 4\alpha, 5\alpha, 6\alpha, 7\alpha$ on the circle (use only decimal parts).

Step 3: Which two points land in the same interval?

Step 4: Their difference = $k\alpha$. Verify $|k\alpha - \text{nearest integer}| < 1/7$.

Step 5: Confirm $1 \leq k \leq 7$.

You have just run the complete Dirichlet proof on a specific case. That is the proof.

Problem 2.3 — The Breaking Point

$$\alpha = 0.5 \text{ exactly}, N = 4.$$

Run the same procedure as Problem 2.2.

What goes wrong if you use closed intervals $[0, 1/4], [1/4, 1/2], [1/2, 3/4], [3/4, 1]$ instead of half-open ones?

This is why Po-Shen Loh was careful about the brackets $[$ and $)$. Feel exactly why.

Hint: where does 0.5 land? Is it in exactly one interval, or on the boundary of two?

BLOCK 3 — Pigeonhole Principle Raw

The principle completely naked. Identify pigeons. Identify holes. State the guarantee.

Problem 3.1 — The Simplest Version

You have 8 people in a room.

Prove at least two of them were born in the same month.

Write this as a formal proof — three lines maximum.

Explicitly state: What are the pigeons? What are the holes? What does PHP guarantee?

Problem 3.2 — Less Obvious

You have 11 integers chosen from $\{1, 2, 3, \dots, 20\}$.

Prove that at least two of them must sum to 21.

Hint: Draw the pairs $\{1, 20\}$, $\{2, 19\}$, $\{3, 18\}$, ... on your slate first.

Then count: how many such pairs are there? How many integers do you have?

Identify pigeons and holes, then apply PHP.

Problem 3.3 — Your DICGC Problem — Prove It By Induction

You discovered the formula $2^n - 1$ for bank accounts. Now prove it formally.

Base case: $n = 1$ person. How many distinct accounts? Verify $2^1 - 1 = 1$. ✓

Inductive step: Assume for n people there are $2^n - 1$ accounts. Add one new person. How many new accounts does this person create with each existing subset?

Conclude: Show the total for $n+1$ people equals $2^{n+1} - 1$.

Write this on your slate. Erase it. Write it again without looking. This is your first formal proof on something you discovered yourself.

BLOCK 4 — Monotone Subsequences

Finding order inside chaos. The theorem says it must be there. Your job is to find it — and build cases where it barely exists.

Problem 4.1 — The Basic Hunt

Given this sequence:

5, 3, 8, 1, 9, 2, 7, 4, 6

$n = 9$, so $\sqrt{9} = 3$. The theorem guarantees a monotone subsequence of length ≥ 3 .

Find by hand:

- The longest monotone increasing subsequence
- The longest monotone decreasing subsequence
- Is either longer than $\sqrt{9} = 3$?

Remember: elements do not need to be adjacent. You can skip over numbers.

Problem 4.2 — Build the Worst Case

$n = 9$ (a perfect square). $\sqrt{9} = 3$.

Construct a sequence of 9 distinct integers where every monotone subsequence — both increasing and decreasing — has length at most 3.

Use the stairs model from the lecture:

Draw $\sqrt{n} = 3$ descending groups of $\sqrt{n} = 3$ elements on your slate. Read the sequence left to right.

Verify your construction works — check both increasing and decreasing directions.

This construction shows $\sqrt{n}$ is tight — you cannot always do better.

Problem 4.3 — The Extremal Case

$n = 16$. $\sqrt{16} = 4$.

Part A: Build a sequence of 16 distinct integers where no monotone subsequence exceeds length 4.

Part B: In your own construction, find a monotone subsequence of length exactly 4 — both an increasing one and a decreasing one.

The theorem says length 4 must exist. Find it in the very sequence you built to prevent long ones.

This is the heart of extremal combinatorics — the tension between construction and guarantee.

BLOCK 5 — The Connection Hunt

Your strongest natural mode. Use it. These go in your Connection Journal — not your slate.

Problem 5.1 — Sequences as Graphs

The lecture connected Erdős to graph theory via the Erdős number.

You connected graph theory to combinatorics in your DICGC insight.

Now: can you model a sequence of n numbers as a graph?

- What would the nodes be?
- What would the edges be?
- Does a monotone subsequence become a known graph concept?

Write whatever you see — even half-formed. Wrong connections are still connections.

Problem 5.2 — Your Own Pigeonhole Moment

Binary search was your instinct in the salary guessing game.

Dijkstra was your childhood globe memory.

Where have you unconsciously used the Pigeonhole Principle in real life before knowing its name?

Think about: scheduling, fitting things into spaces, distributing anything among limited containers.

Write one real example from your life. Then write the formal pigeonhole statement that describes it.

This is the Ramanujan move — recognizing the structure you already knew, now with its name.

END-OF-DAY RITUAL — Every Night This Week

Before sleeping, write these three things in your Connection Journal. Not on the slate — this record is permanent.

1. The Core Essence	Complete this sentence in your own words: "Dirichlet's theorem is really saying that..." Not the formal statement. The feeling of what it means.
2. The Open Question	The one thing that still feels uncertain after today. Write it as one clear sentence. Plant it before sleeping. Your unconscious mind will work on it tonight.
3. The Connection	One link between today's material and anything else you know — another subject, real life, childhood, etc. Even a weak or uncertain connection counts. Write it.

"What I cannot create, I do not understand."
— Richard Feynman

Your intuition already works like Ramanujan. Now build the proof language to match it.